Rerandomization simulation report

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Introduction

In the course of our empirical work, we found a surprising range in both point estimates and, especially, standard errors. To better understand this variability, we ran a series of simulations calibrated to the structure of the data we were analyzing. These simulations allowed us to assess whether such ranges might arise naturally, even in the absence of treatment effects.

They also gave us a way to evaluate the calibration of standard errors—specifically, whether the estimated standard errors correspond to the actual variation in point estimates across permutations.

We generated a series of research questions, listed below. To answer these, we generated synthetic datasets with no treatment effect and examined the distribution of resulting estimates across permutations of treatment assignment. This approach preserved the structure of the data, including covariate relationships and cluster sizes, while removing any actual treatment effect. By looking at the estimates across the simulated datasets, we can assess whether the empirical findings could be expected under a scenario of no treatment variation.

We next give an overview of the RQs with top-level findings, and then dive into the analysis giving evidence for these claims.

RQ: Are the ranges of point estimates and standard errors across methods in the empirical data consistent with what we'd expect under no treatment effect?

Yes. Our simulation gives ranges that mirror the ranges we saw empirically.

RQ: Are the estimated standard errors well calibrated—that is, do they reflect the true variability in estimates?

In general, the estimated SEs are about right. There are some notable exceptions, especially for the matched-pairs experiment, where some methods have estimated SEs that are very much too low.

The permutation style simulation does not allow a good assessment of how much the estimated SEs change across datasets; a second simulation should be done to investigate estimated SE stability.

RQ: Are some estimators more precise than others in some contexts?

We do see some evidence that, for some datasets, some estimators are up to 30% more variable than other estimators. For other datasets, the estimators seem to be about the same.

These differences are much more modest than the apparent differences when we compare the *estimated* Standard Errors.

RQ: Are some of the standard errors imprecisely estimated – i.e., do some methods have standard errors that are unstable, giving extreme values or incorrect values much of the time?

Yes, some methods have estimated standard errors that are very unstable, giving extreme values or incorrect values much of the time. This is especially true for the matched-pairs experiment, where some methods have

estimated SEs that are very much too low.

But we need simulation to see if it is systematically too low, vs systematically off (sometimes high, sometimes low, depending on the dataset).

RQ: Do the empirical results appear unusually variable, suggesting something atypical in the data?

The empirical results seem to be in line with what we are seeing in the simulation.

The process of the rerandomization simulation

To generate simulation data I did the following for each outcome j in our set of outcomes across our empirical dataset (excepting a few studies that were inaccessible):

1) Load the empirical data for outcome j, patch missing data, etc.
2) Calculate the empirical estimates of the treatment effects for outcome j, and the associated empirical standard errors.
3) Repeatedly, for simulation runs $r = 1, \ldots, R$ (with $R = 20$), permute the treatment assignment across clusters and within blocks. I then re-estimated the impacts given the permuted assignment.
4) I then, within each run and outcome, calculated the range of impact estimates, and the ratio of the largest to smallest standard error, across all the estimators that were defined for that run. I did this both overall, and for the group of estimators within each estimand (Person or Cluster).
5) I finally saved the set of results to a file.

Due to the permutation, there is no actual average treatment effect or cross-block treatment effect variation. All individuals have the same outcome whether assigned to treatment or control. Any actual treatment effect in the empirical data is therefore pushed into individual and cluster variation. For example, if a cluster had a positive average effect, the permutation simulation views this cluster as just having a slightly higher intercept than it would have had, had it been a control cluster in reality. The structure of the data, the relationship of covariates to outcome, etc., is all preserved by this approach.

The datasets missing from this analysis

We didn't run the simulation on all the empirical datasets due to privacy restrictions and computational feasibility. The datasets dropped are the following:

```
##
##
## Table: Dropped Datasets
##
## |UniqueID                 |      n|   J|
## |:------------------------|------:|---:|
## |CARES_DC_PT_Incred       |   1298|  52|
## |CARES_TR_BPI_Incred      |   1318|  52|
## |CARES_TR_WRS_Incred      |   1320|  52|
## |CARES_DC_EMIT_PTH        |   1229|  52|
## |CARES_DC_CST_COMPTOT_PTH |   1206|  52|
## |CARES_TR_SSRS_PTH        |   1276|  52|
## |CARES_DC_PT_Tools_Mind   |   1268|  52|
## |CARES_TR_WRS_Tools_Mind  |   1293|  52|
## |CLC_GRCCSS_POST          |   9557|  28|
## |CLC_PCCRERCORE           |  16465|  28|
## |MathPD_MTSS              |   2132|  39|
## |MathPD_MTSS_FRACTDEC     |   2132|  39|
## |MathPD_MTSS_PCTRATIO     |   2132|  39|
```

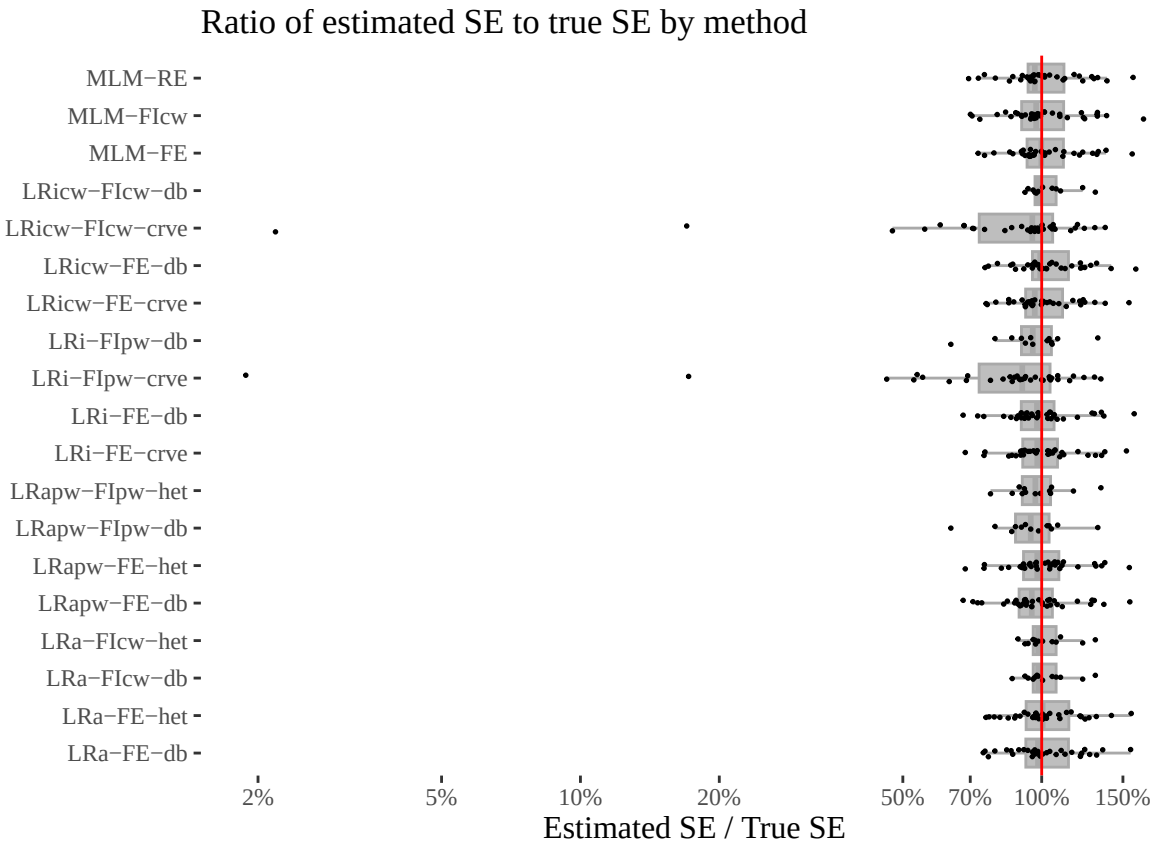
```
## |jg_crt_6                 |  15993| 802|
## |jg_crt_52                | 102869| 508|
```

TODO: Cross check the missing simulations to empirical results. Are any of the dropped data sets the ones that were problematic?

How do the simulation estimated SEs compare to the actual simulation SE?

For each scenario, we can calculate true standard errors (SEs) for each method by taking the standard deviation of the point estimates across all permutations. Each simulation iteration also gives an estimated standard error along with point estimate for each method. We can then check to see if the average of the estimated standard errors (across iterations) correspond to the true standard error. This tells us whether our estimated standard errors are accurately capturing uncertainty–in this case capturing finite sample uncertainty given the fixed data and only letting the treatment assignment vary.

In short, within simulation scenario and for each method, we can compare the average estimated SE to the standard deviation of point estimates across all permutations. In the plot below we show the distribution of ratios of $E[\widehat{SE}]/SE$ across the studies, for each method.



Some methods do not have standard errors for some scenarios (if there are too few degrees of freedom or there are too few clusters in some blocks), so they are not in the above plot–note how some boxes have fewer points. For many methods, the average estimated SEs are generally close (well within 50%) of the true SEs.

Note the FI CRVE outliers: cluster robust standard errors can give extremely low values in some contexts. We apparently see it here. Our simulation should be able to show us whether this happens regularly or not in the context we are exploring.

For the record, the outlier points that are 50% or less than expected are the following:

method	UniqueID	SE	ESEhat	sdSEhat
LRi-FIpw-crve	TRIADScale-up_rema2	0.27	0.01	0
LRi-FIpw-crve	jg_crt_31	0.04	0.02	0
LRi-FIpw-crve	jg_crt_63	0.03	0.01	0
LRicw-FIcw-crve	TRIADScale-up_rema2	0.26	0.01	0
LRicw-FIcw-crve	jg_crt_31	0.04	0.02	0
LRicw-FIcw-crve	jg_crt_63	0.03	0.01	0

Some methods do not provide estimated standard errors for some of the study designs explored. Counts of the methods with and without standard errors are below.

```
## # A tibble: 19 x 2
##    method          num_emp
##    <chr>             <int>
##  1 LRa-FE-db            35
##  2 LRa-FE-het           35
##  3 LRa-FIcw-db          34
##  4 LRa-FIcw-het         34
##  5 LRapw-FE-db          35
##  6 LRapw-FE-het         35
##  7 LRapw-FIpw-db        34
##  8 LRapw-FIpw-het       34
##  9 LRi-FE-crve          35
## 10 LRi-FE-db            35
## 11 LRi-FIpw-crve        35
## 12 LRi-FIpw-db          34
## 13 LRicw-FE-crve        35
## 14 LRicw-FE-db          35
## 15 LRicw-FIcw-crve      35
## 16 LRicw-FIcw-db        34
## 17 MLM-FE               35
## 18 MLM-FIcw             35
## 19 MLM-RE               35
```

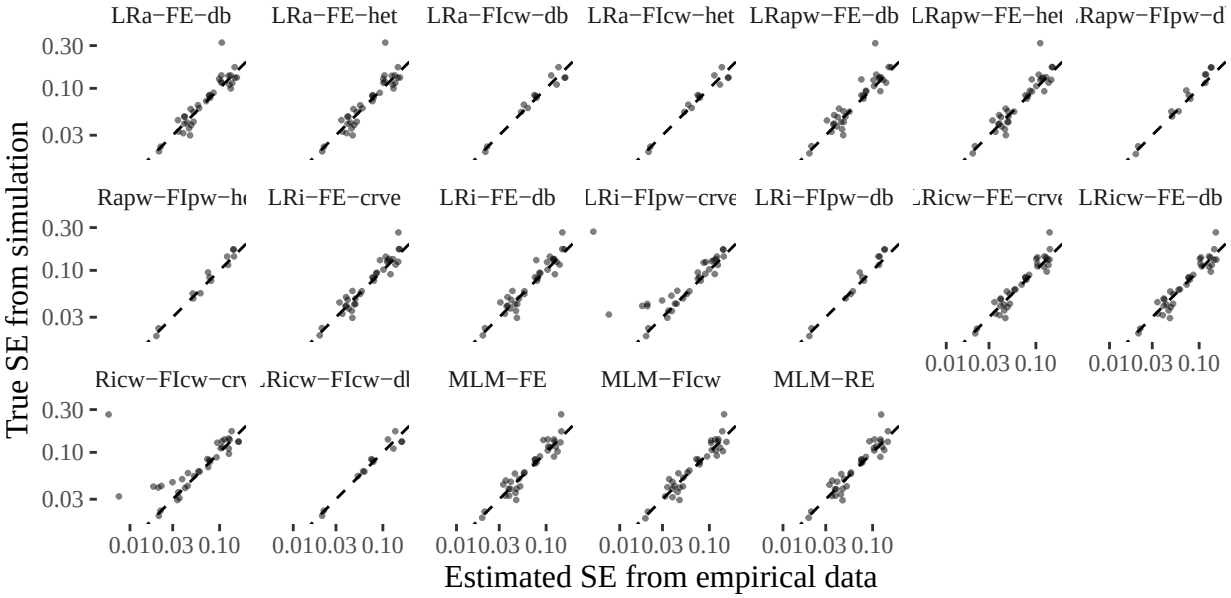
method	n	SE_def	ESEhat_def	num_emp
LRa-FE-db	35	35	35	35
LRa-FE-het	35	35	35	35
LRapw-FE-db	35	35	35	35
LRapw-FE-het	35	35	35	35
LRi-FE-crve	35	35	35	35
LRi-FE-db	35	35	35	35
LRi-FIpw-crve	35	35	35	35
LRicw-FE-crve	35	35	35	35
LRicw-FE-db	35	35	35	35
LRicw-FIcw-crve	35	35	35	35
MLM-FE	35	35	35	35
MLM-FIcw	35	35	35	35
MLM-RE	35	35	35	35
LRa-FIcw-db	35	34	13	34
LRa-FIcw-het	35	34	13	34

method	n	SE_def	ESEhat_def	num_emp
LRapw-FIpw-db	35	34	13	34
LRapw-FIpw-het	35	34	13	34
LRi-FIpw-db	35	34	13	34
LRicw-FIcw-db	35	34	13	34

Here `num_emp` is the number of empirical (original) defined ATE estimates across the set of studies.

How do the estimated SEs for the real data compare to the true simulation SEs?

For the most part, the average estimated SEs tend to be fairly close to the estimated *true* SEs we obtained via our permutation simulation. Below, we compare the estimates we got on the original empirical data to these permutaton-based true SEs.



We do see some outliers, where the true SE is way higher or way lower than what we estimated. We filter to those studies where this occurred, showing all studies where the estimated SE is more than 50% larger than the true SE or less than 70% of the true SE:

Collection	UniqueID	method	SE_hat	SE	off_by
Curated	TRIADScale-up_rema2	LRi-FIpw-crve	0.01	0.27	53.15
Curated	TRIADScale-up_rema2	LRicw-FIcw-crve	0.01	0.26	45.82
Public	jg_crt_63	LRicw-FIcw-crve	0.01	0.03	4.28
Public	jg_crt_63	LRi-FIpw-crve	0.01	0.03	4.24
Curated	TRIADScale-up_rema2	LRa-FE-db	0.11	0.32	3.06
Curated	TRIADScale-up_rema2	LRa-FE-het	0.11	0.32	3.04
Curated	TRIADScale-up_rema2	LRapw-FE-db	0.11	0.32	2.98
Curated	TRIADScale-up_rema2	LRapw-FE-het	0.11	0.32	2.85
Public	jg_crt_31	LRicw-FIcw-crve	0.02	0.04	2.28
Public	jg_crt_31	LRi-FIpw-crve	0.02	0.04	2.25
Curated	TRIADScale-up_rema2	MLM-RE	0.12	0.26	2.12
Public	jg_crt_68	LRi-FIpw-crve	0.02	0.04	2.08

Collection	UniqueID	method	SE_hat	SE	off_by
Public	jg_crt_69	LRi-FIpw-crve	0.02	0.04	1.96
Public	jg_crt_69	LRicw-FIcw-crve	0.02	0.04	1.95
Public	jg_crt_68	LRicw-FIcw-crve	0.02	0.04	1.88
Curated	TRIADScale-up_rema2	LRicw-FE-crve	0.14	0.27	1.86
Curated	TRIADScale-up_rema2	MLM-FIcw	0.15	0.27	1.82
Curated	TRIADScale-up_rema2	MLM-FE	0.15	0.27	1.80
Curated	TRIADScale-up_rema2	LRi-FE-crve	0.15	0.26	1.77
Curated	TRIADScale-up_rema2	LRi-FE-db	0.15	0.26	1.74
Curated	TRIADScale-up_rema2	LRicw-FE-db	0.15	0.27	1.73
Curated	TRIAD_rema2	LRapw-FE-db	0.07	0.13	1.68
Curated	TRIAD_rema2	LRi-FE-db	0.08	0.13	1.67
Curated	Beacon_S8TOTCR	LRicw-FIcw-crve	0.03	0.05	1.56
Curated	Beacon_S8TOTCR	LRi-FIpw-crve	0.03	0.05	1.55
Curated	Beacon_S8TGPA	LRapw-FE-het	0.05	0.03	0.65
Curated	Beacon_S8TGPA	LRa-FE-het	0.05	0.03	0.65
Curated	Beacon_S8TGPA	LRi-FE-crve	0.05	0.03	0.65
Curated	Beacon_S8TGPA	LRicw-FE-crve	0.05	0.03	0.64
Curated	Beacon_S8TGPA	LRapw-FE-db	0.05	0.03	0.64
Curated	Beacon_S8TGPA	LRa-FE-db	0.05	0.03	0.64
Curated	Beacon_S8TGPA	MLM-FE	0.05	0.03	0.63
Curated	Beacon_S8TGPA	MLM-FIcw	0.05	0.03	0.63
Curated	Beacon_S8TGPA	MLM-RE	0.05	0.03	0.63
Curated	Beacon_S8TGPA	LRi-FE-db	0.05	0.03	0.62
Curated	Beacon_S8TGPA	LRicw-FE-db	0.05	0.03	0.62

`off_by` is $SE/\widehat{SE}$.

Some stats on the studies that are responsible:

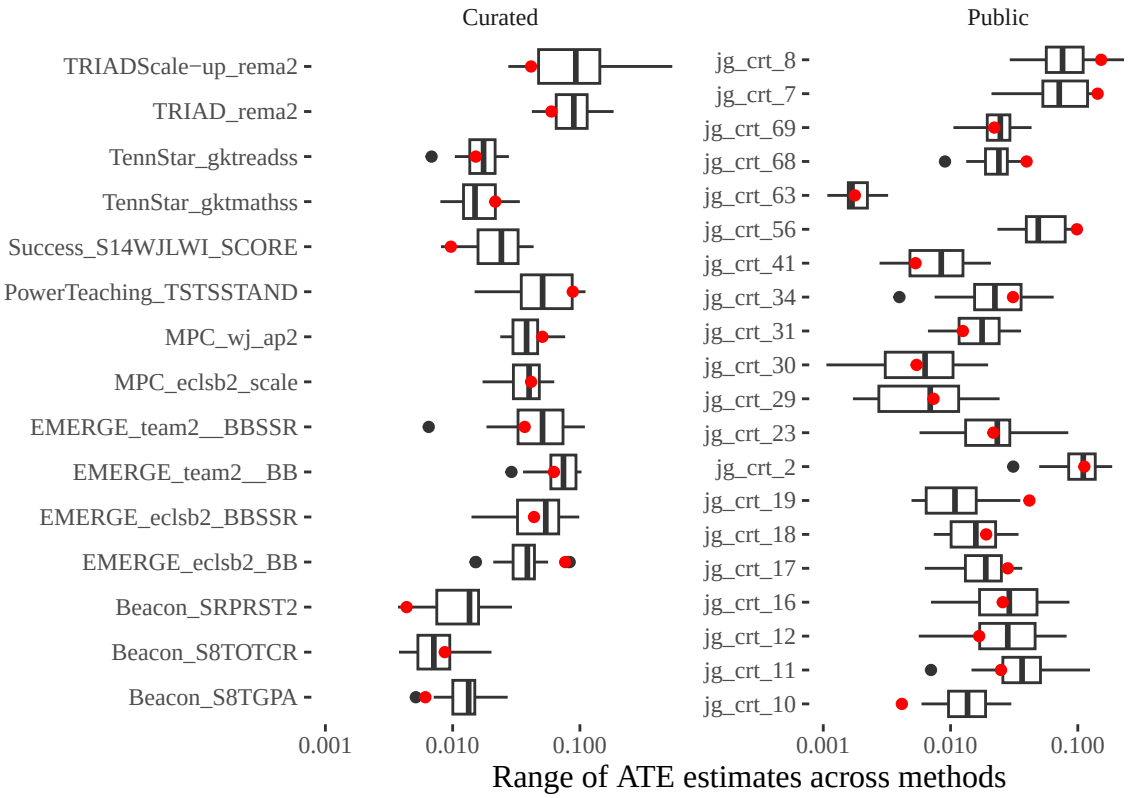
UniqueID	Collection	num_show	n	J	Jcv	Jbar	cons_df
Beacon_S8TGPA	Curated	11	2011	83	0.94	3.61	37
Beacon_S8TOTCR	Curated	2	2165	83	0.94	3.61	37
TRIADScale-up_rema2	Curated	13	409	16	0.00	2.00	0
TRIAD_rema2	Curated	2	1208	38	0.10	5.43	24
jg_crt_31	Public	2	11395	381	0.18	4.33	205
jg_crt_63	Public	2	8987	179	0.15	2.11	9
jg_crt_68	Public	2	8761	381	0.18	4.33	205
jg_crt_69	Public	2	8758	381	0.18	4.33	205

For the matched pairs design (TRIAD Scale Up), we often got an estimated SE that was way too small. `jg_crt_63` is almost a matched pairs experiment (Jbar of 2.11), and shows similar issues.

TODO: Explore why 68 and 31, listed above, are all messed up. Are they messed up in other ways?

How do the actual ranges we saw compare to the ranges we get from the simulation?

For each study, we can look at the actual range of ATEs from the empirical data (red dot) vs. the ranges we get from the permutations (the boxplot). The boxplot shows what kinds of ranges we would "naturally" see when there is no treatment effect or treatment variation.

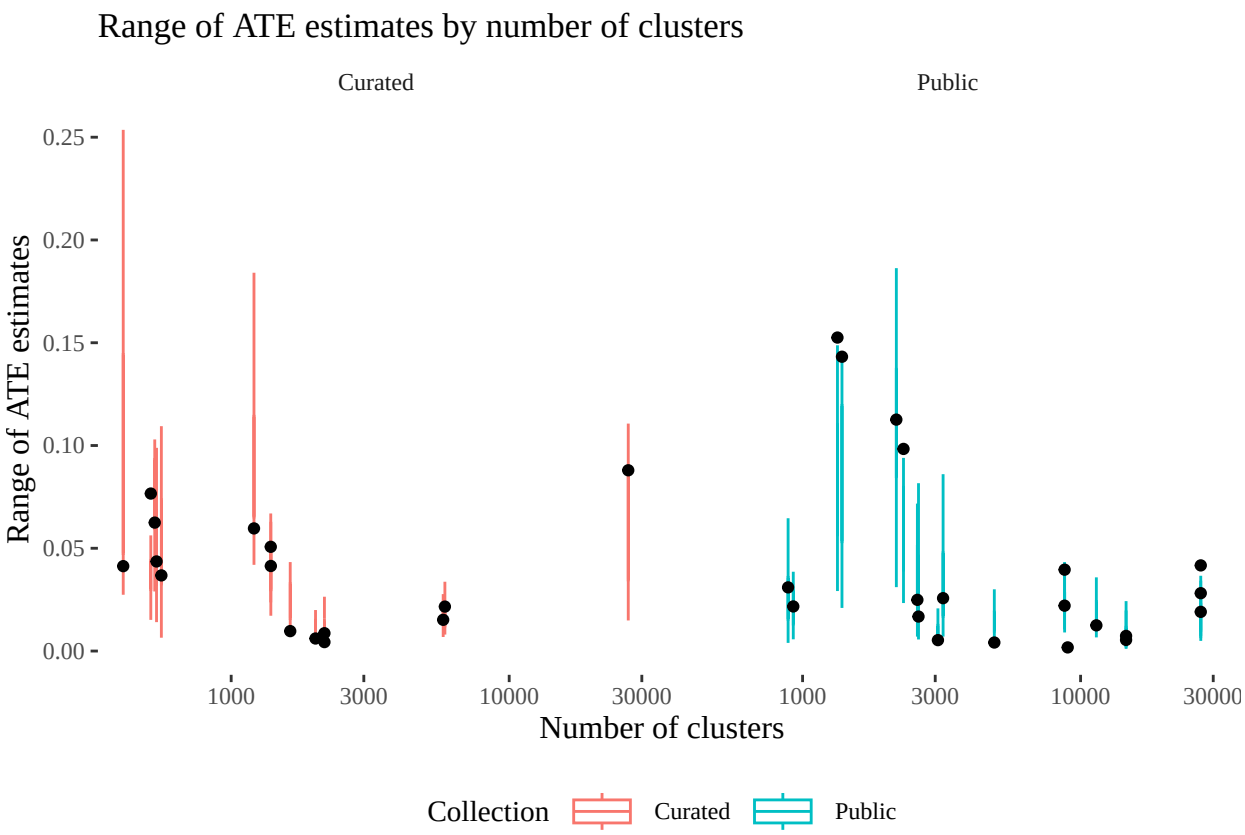


The red dots generally lie in the range of the boxplots; this means the range we saw in our empirical data is entirely consistent with the ranges we would see if there were no treatment effect, or variation in treatment effect, whatsoever. Our observed range, therefore, does not appear to be driven by the estimators targeting different estimands.

To quantify this, we can calculate the percent of studies where the empirical range is outside the 90% interval of the simulated ranges:

Collection	per_below_05	per_above_95	percent_outside_90
Curated	7	0	7
Public	5	15	20
Overall	6	9	14

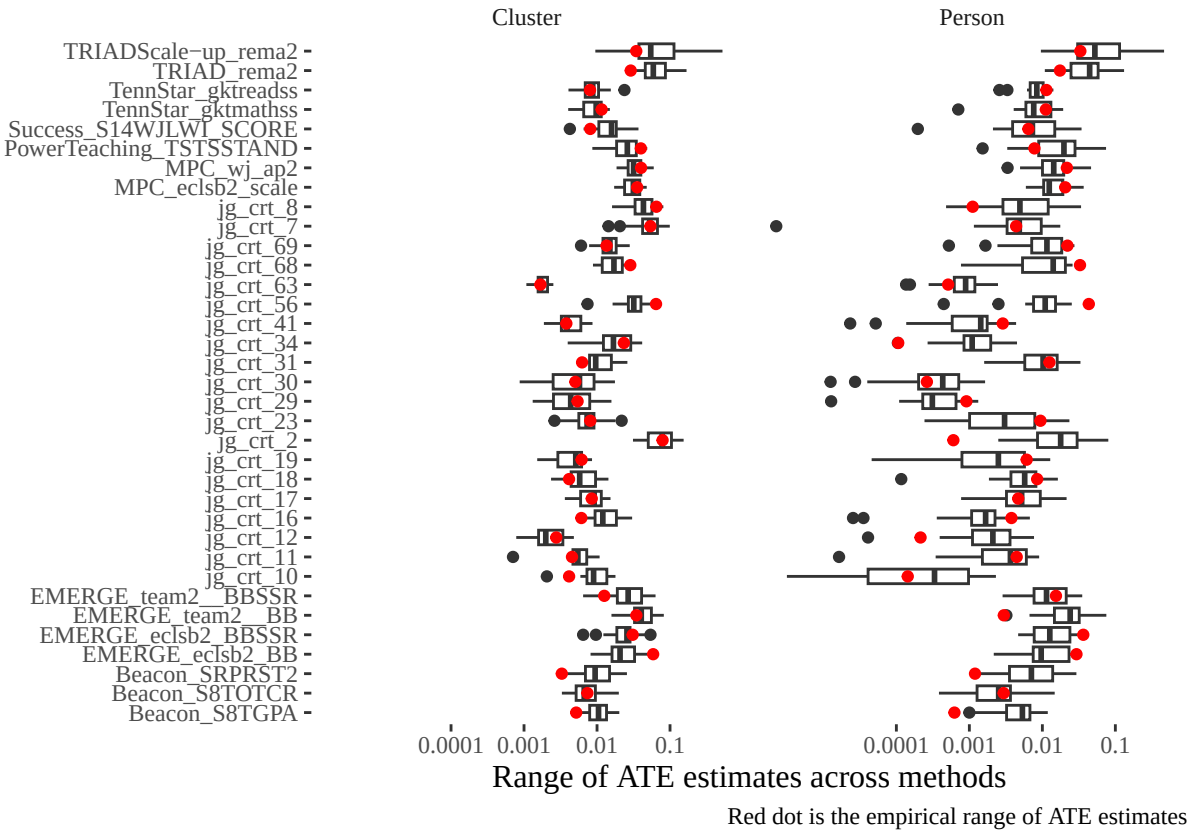
We can look at the range of ATE estimates vs size of study (number of clusters). The empirical are plotted as dots on the lines.



Note: When there are multiple outcomes for the same study, these lines are overplotted.

Within Estimand

We can look at the ranges within estimand, replicating the above.

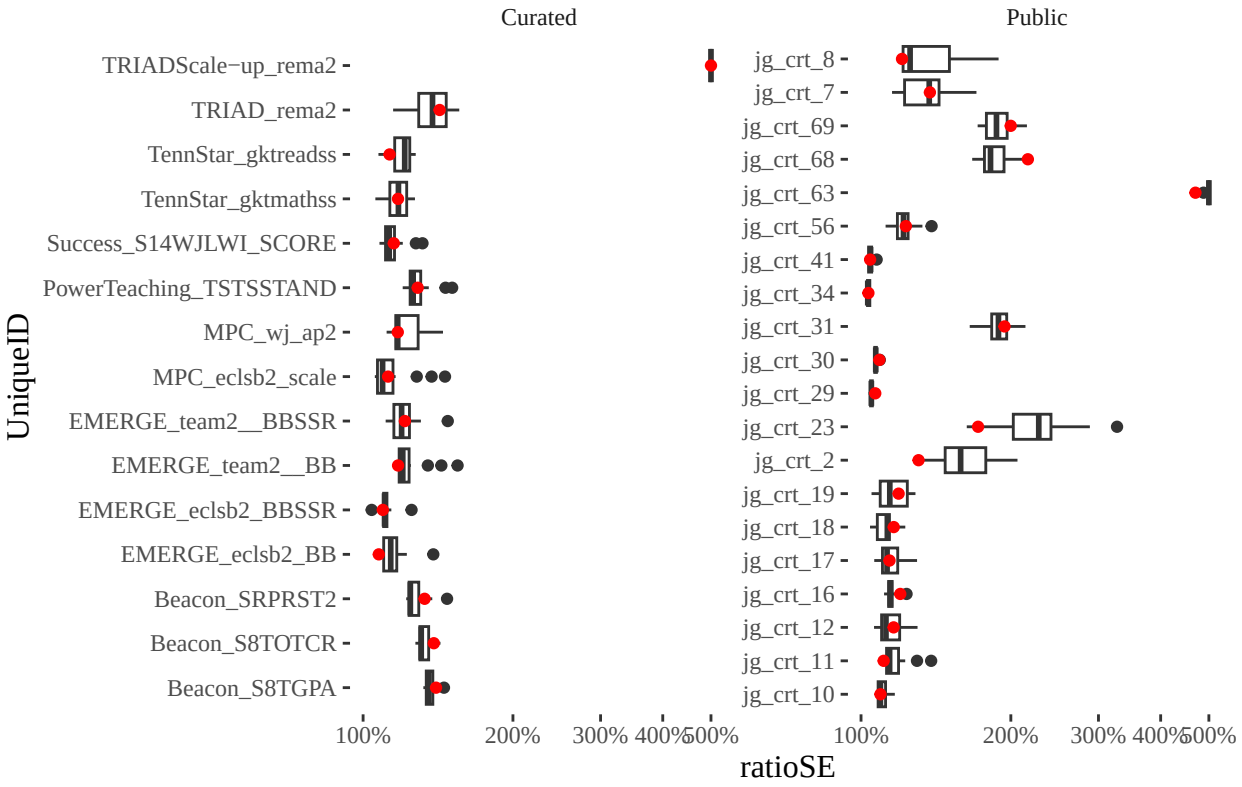


Again things look relatively consistent with our empirical lining up with what we see in the simulation.

How do the actual SE-hat ratios compare to what we get from simulation?

We next look at how much the ratio of the max to min estimated standard errors vary across the permutations, and compare that to the ratios from the empirical data.

TODO: Look at the studies with big ranges and big estimated ranges. Do they correspond to the empirical results and identified studies in the "flashcard report"? Do a cross check to see if the "weird studies" there are captured here, and vice-versa. (We noticed some discrepancies and need to follow up.)



For most studies, we the ratio of high to low SE is fairly consistant, with some studies having high ratios and some low. TRIAD, for example, always had ratios of 5 or more. `jg_crt_31` always had ratios of around 2. Other studies had smaller ratios, close to 1, meaning all the estimators generally gave the same uncertainty estimate for each iteration.

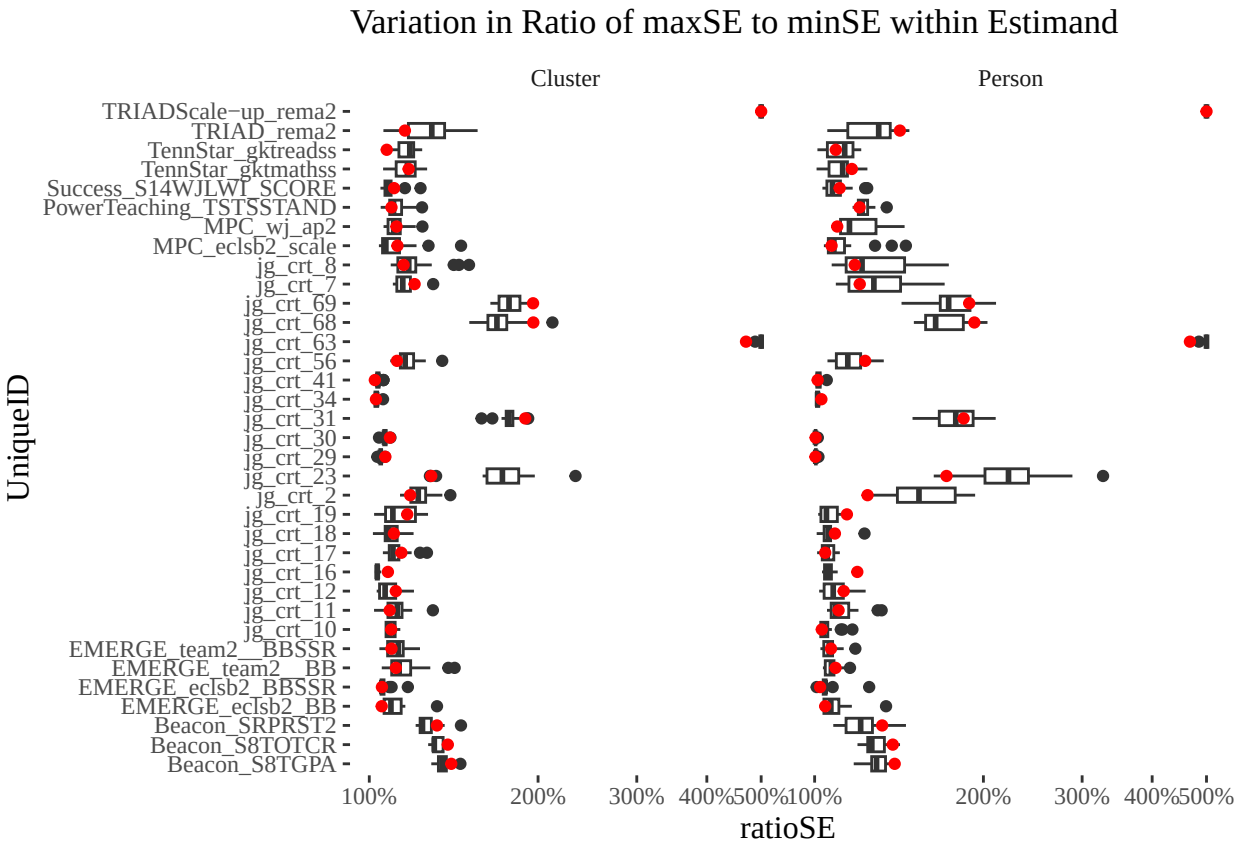
Collection	per_below_05	per_above_95	percent_outside_90
Curated	7	0	7
Public	10	25	35
Overall	9	14	23

Overall, we are seeing more empirical values in the tails of the ranges–overall 23% of the studies have ratios in the 5% tails (10% total) of the simulated ranges.

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Within Estimand

We can look at the ranges within estimand, replicating the above.

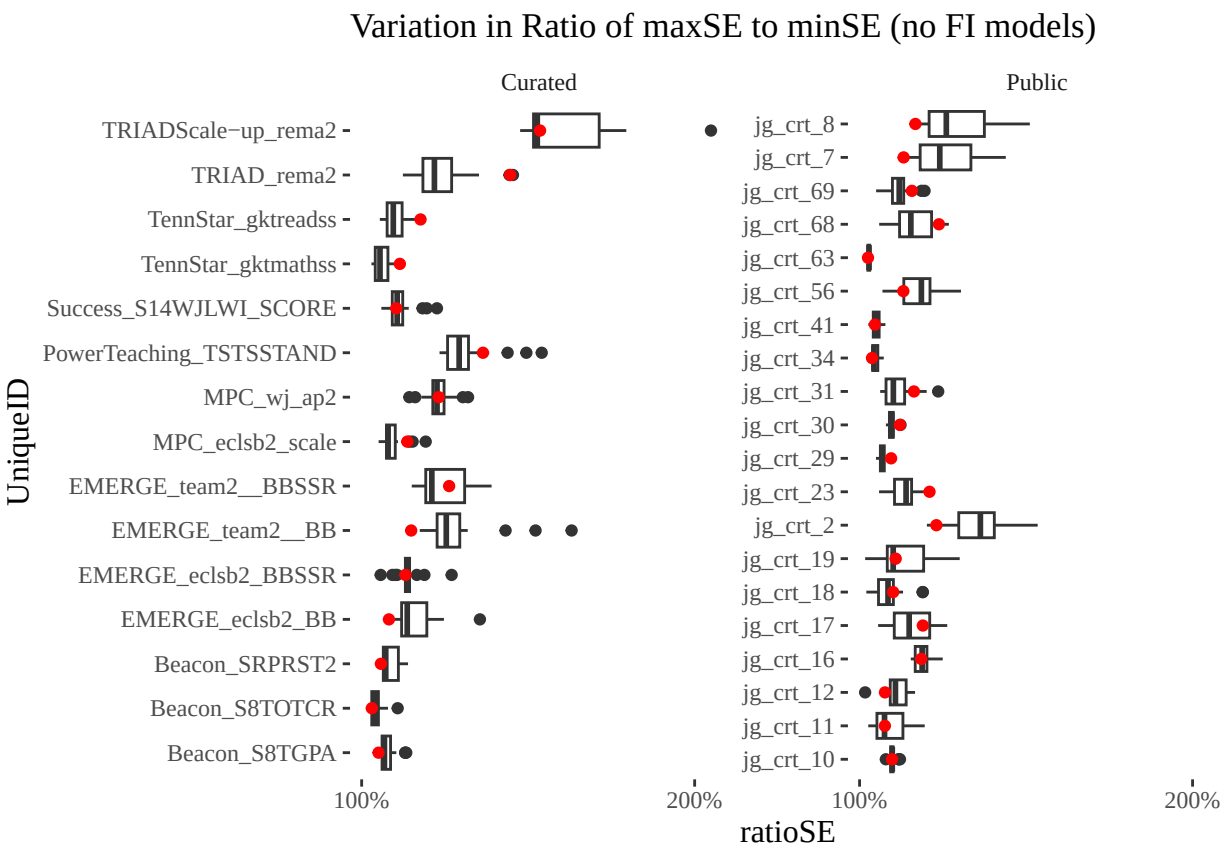


Collection	Estimand	per__below__05	per__above__95	percent__outside__90
Curated	Cluster	0	0	0
Public	Cluster	10	15	25
Overall	Cluster	6	9	14
Curated	Person	0	13	13
Public	Person	10	10	20
Overall	Person	6	11	17

We see again some more extreme empirical values than we would expect from the permutaton.

Without FI models

We can replicate the above, looking at the ratios for the non-FI models only. Again, the observed ranges seem in line with the ones we get from our simulation. The ratios are all notably smaller when we drop FI; this is sensible in that there are fewer estimators to compare, and thus the ratio by definition has to go down (or stay the same at the very worst).

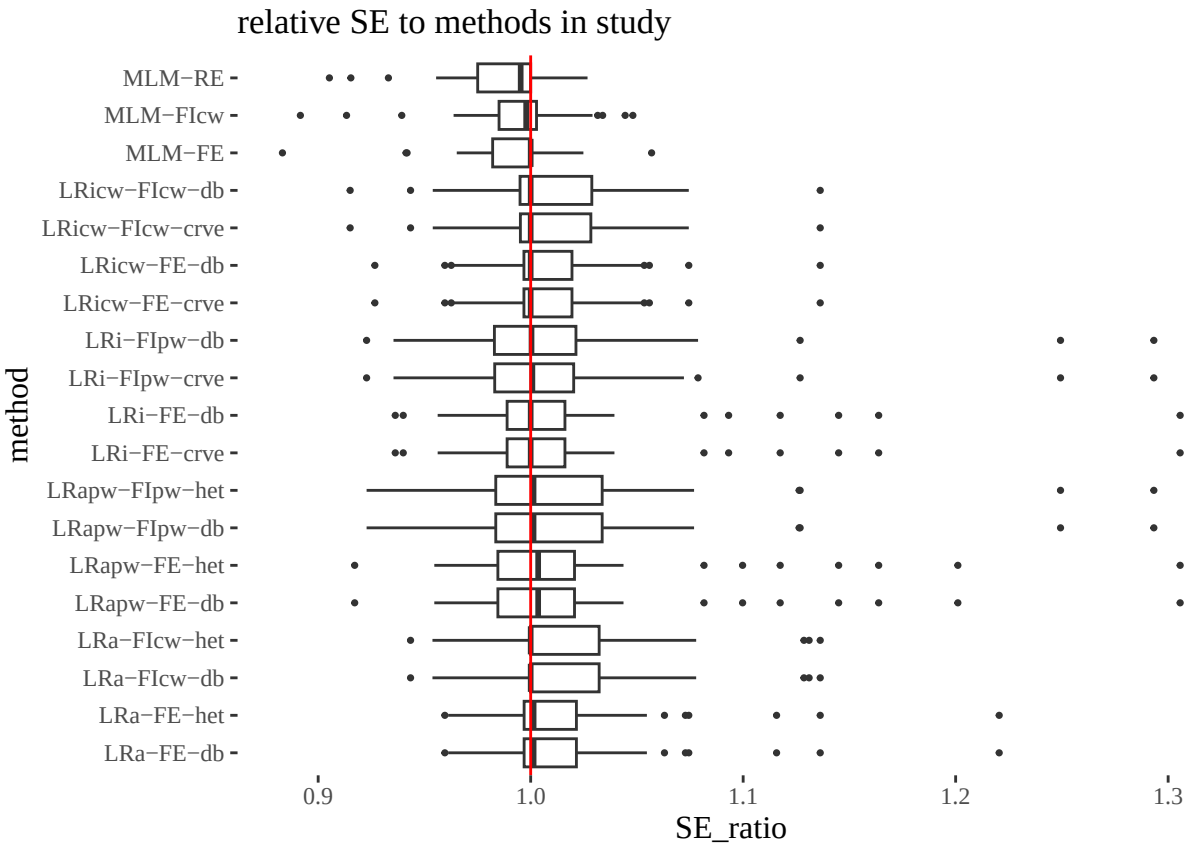


Examining estimator performance

In the following I continue to examine how the different estimators perform, relative to the other estimators.

Question 1: Do some methods have higher true standard errors than other methods?

First we compare standard errors of methods by looking at the ratio of the true SE vs. median true SE within study, and then making a boxplot

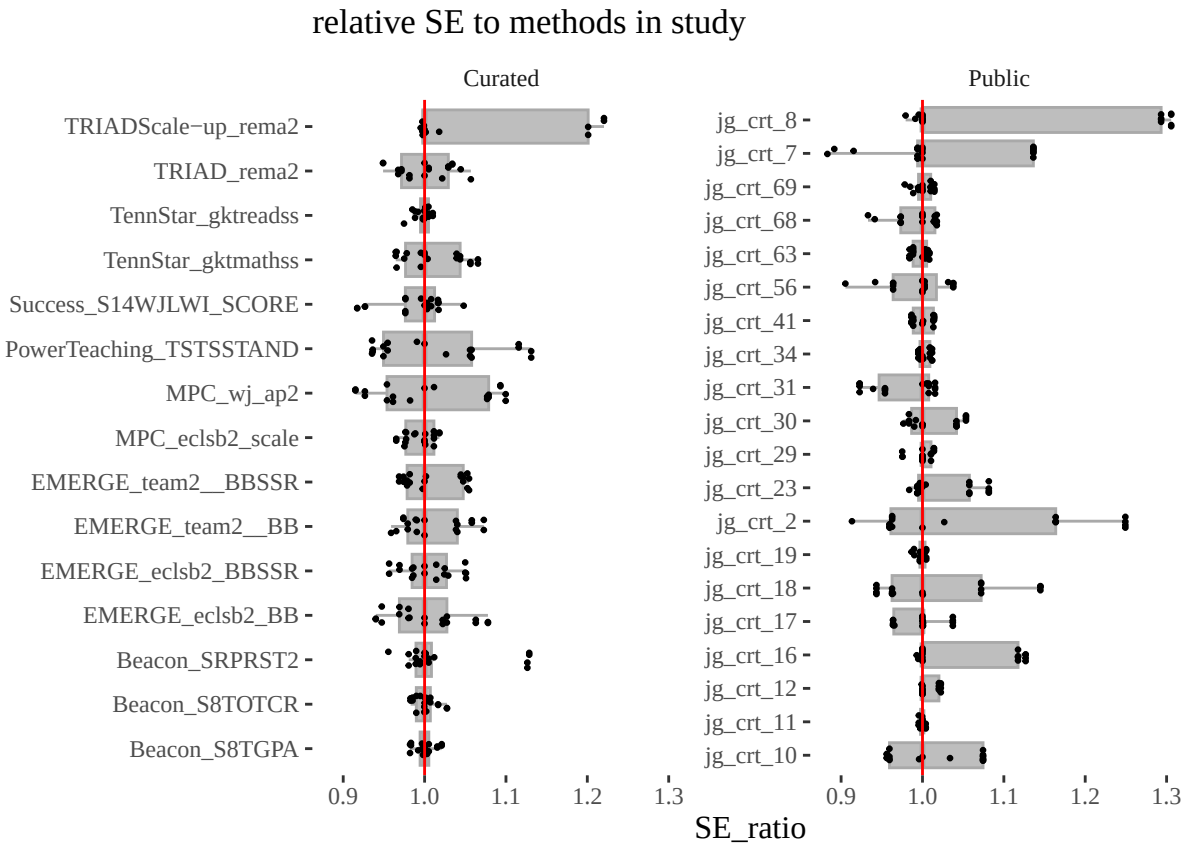


Discussion: All the methods have a SE ratio that is, at the median, the same as the other methods. This suggests that the methods all have the same overall performance across datasets, on average. That said, some methods have long right tails, meaning there are some studies where their true SEs are up to 30% larger than the median performing method.

We can organize these same values by study to see if some studies naturally lead to bigger ranges of true SE:

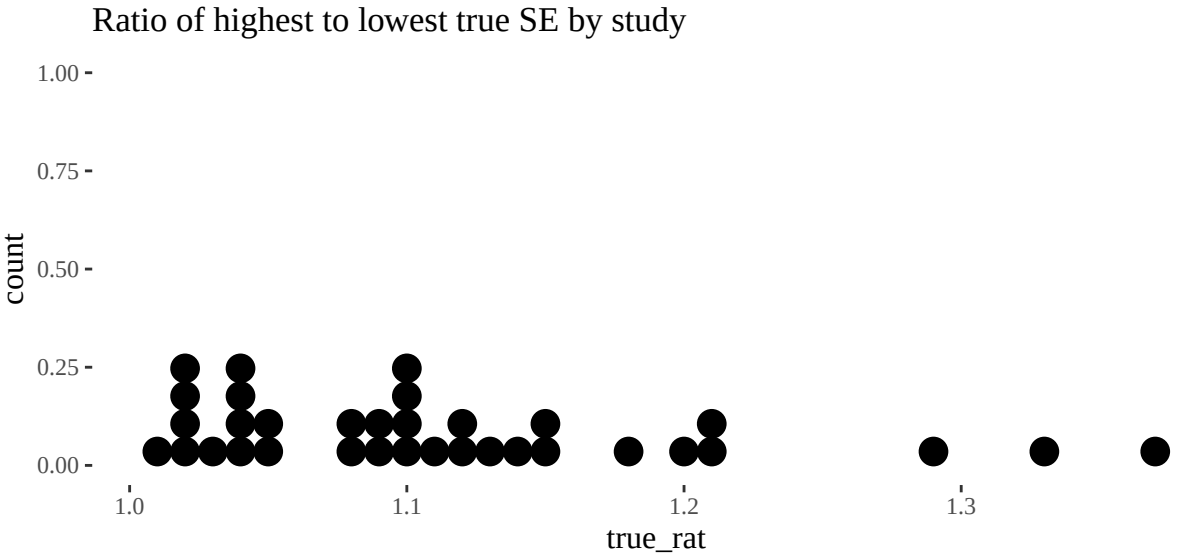
```
## Warning: Removed 6 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



The second plot shows that some studies indeed have a wider range of true SEs than other studies. For example, some estimators in `jg_crt_2` have larger SEs and some have smaller SEs. By contrast, almost everyone in `Beacon_S8TGPA` have the same true SE (note the very narrow boxplot).

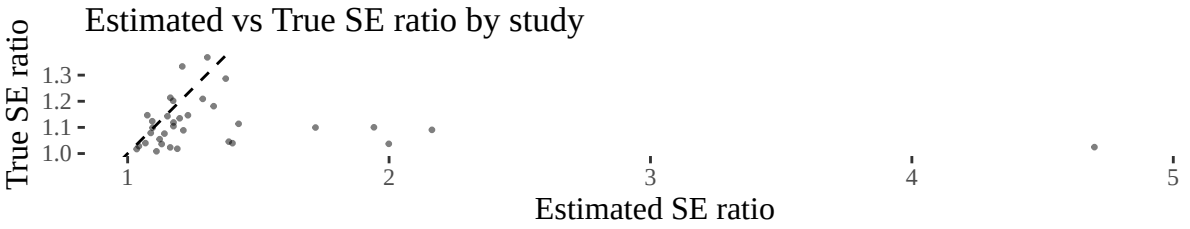
We can look at it like this, with the range of *true* SEs by study. This is the "true" version of our main ratio plot.



These ratios are all inflated by simulation error (we do not fully know the true SEs), but even so, we are not seeing major ratios in the TRUE SE, even though we do see them in the *estimated* SE. The long range we see in the empirical is because the *estimated* SEs vary a lot, not because the estimators are very different in

terms of their precision. We have seen some estimators have very poorly estimated SEs, and this means the ratios are driven high when this happens.

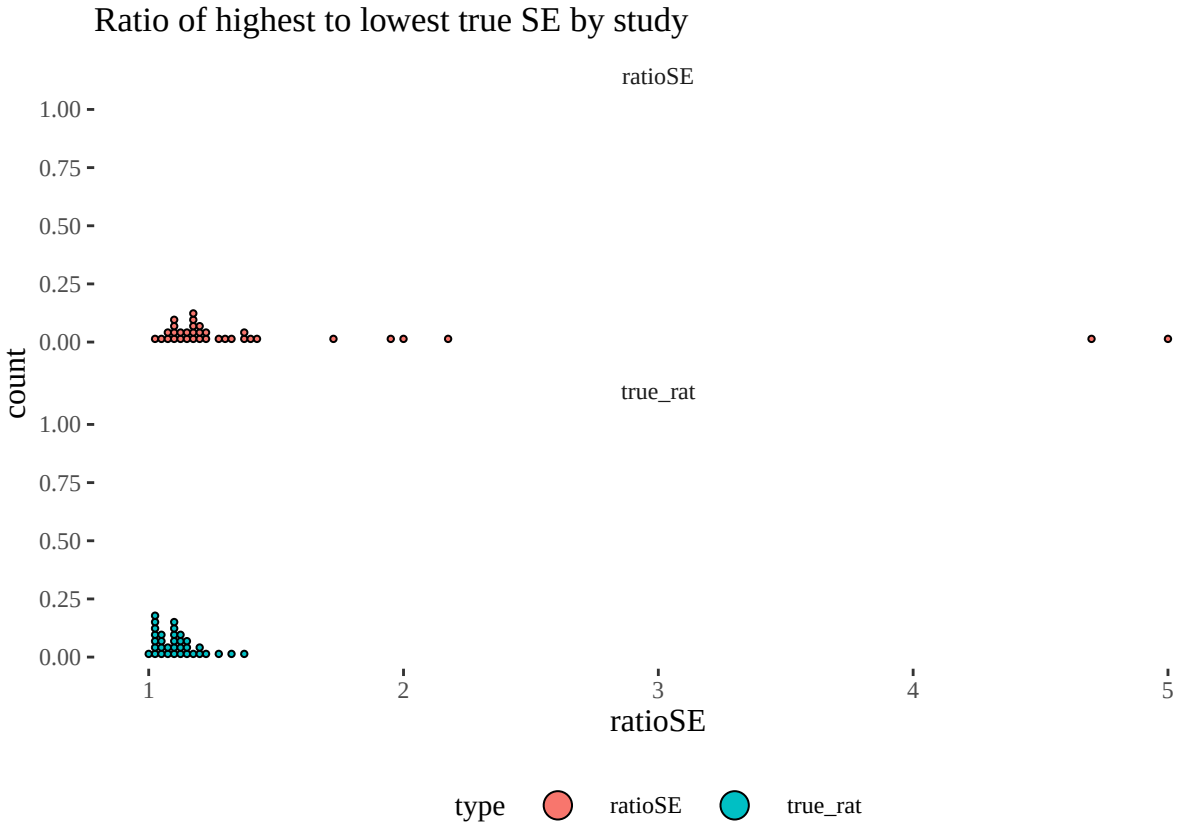
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



The above plot shows how the true ratio is very different from the max/min ratio of the estimated SEs for many of our studies. It also shows how the estimated ratios can be very much larger than the true ones.

We can look at it in light of the figure from the main paper:

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`stat_bindot()`).
```



In the above we see the clear contraction of the ratios when we look at *true* SE ratios vs *estimated* SE ratios.

Note: Looking at the distribution of the ratios of estimated SE under the permutation approach gives basically the exact same picture, and is harder to scale, so I am leaving it as this.

Addendum: Modeling to understand variation NOTE: This part is rough work in progress

The true SEs are estimated with error, across only 20 simulation replicates. We can account for this uncertainty by fitting a model where we regress $log(\widehat{ATE}^2)$ onto a random effect for study, method, and

study by method.

The average error is essentially the log of the estimated SE. Note that

$$SE = E[(\widehat{ATE} - E[ATE])^2].$$

We then plug in $E[ATE]$ to get individual permutation estimates of the standard error:

$$err = (\widehat{ATE} - \frac{1}{R}\sum_{r=1}^{R} \widehat{ATE}_r)^2$$

where R is the number of permutations. Then regressing the log of these onto method and study will let us know if some methods have systematically higher or lower SEs than others, controlling for study. We include interactions to allow these differences to be different for some studies.

```
## boundary (singular) fit: see help('isSingular')
```

```
## lmer(formula = err ~ 1 + (1 | method) + (1 | UniqueID) + (1 |
##     method:UniqueID), data = simdat)
## coef.est  coef.se
##    -6.49     0.22
##
## Error terms:
##  Groups          Name        Std.Dev.
##  method:UniqueID (Intercept) 0.00
##  UniqueID        (Intercept) 1.29
##  method          (Intercept) 0.00
##  Residual                    2.11
## ---
## number of obs: 13839, groups: method:UniqueID, 659; UniqueID, 35; method, 19
## AIC = 60136.2, DIC = 60123.7
## deviance = 60124.9
```

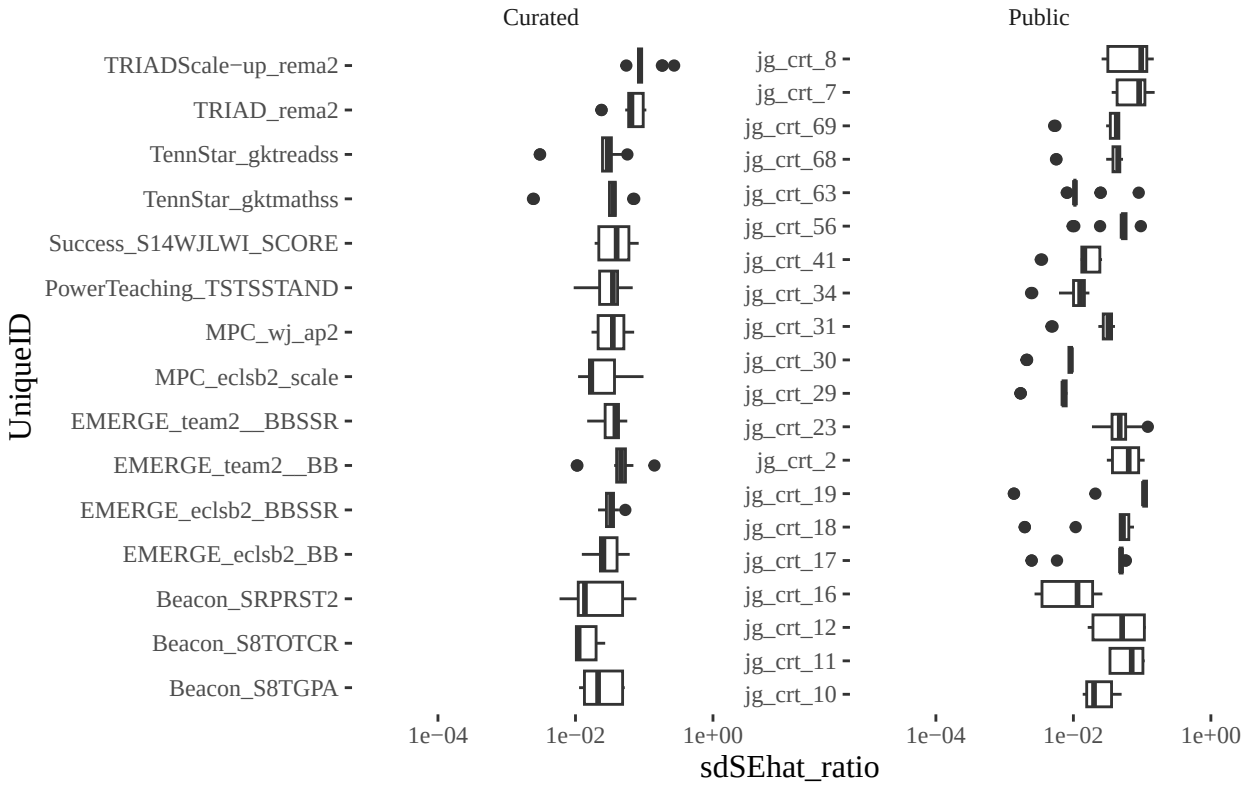
```
##  Groups          Name        Std.Dev.
##  method:UniqueID (Intercept) 0.0000
##  UniqueID        (Intercept) 1.2875
##  method          (Intercept) 0.0000
##  Residual                    2.1108
```

Interestingly, this shows that we have no variation in SE for the different methods. This underscores that whatever the differences in SEs are across methods, they are minimal.

Allowing for heteroskedasticty (not working)

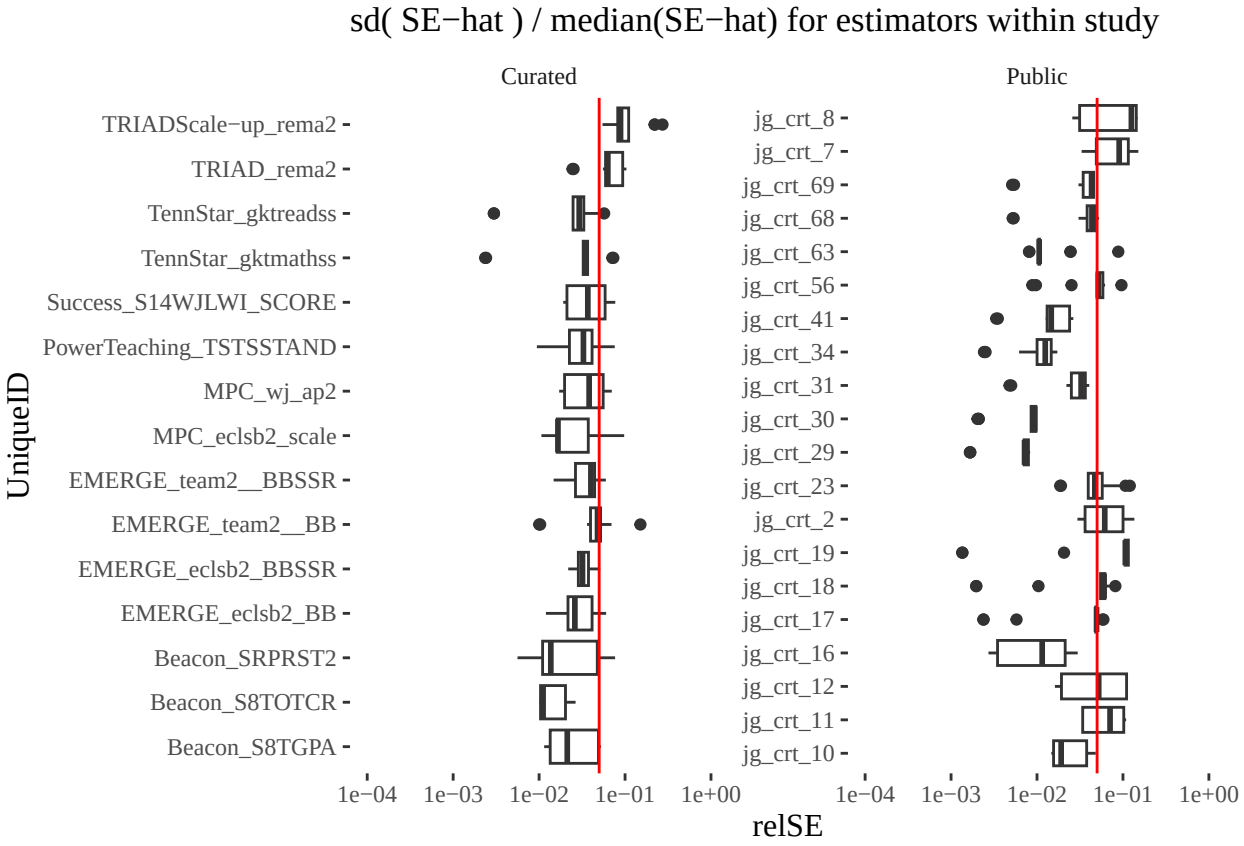
Question 2: Do some methods have more trouble estimating their standard error?

For each estimator in each study, we can calculate $sd(\widehat{SE})/SE$. We then make boxplots of those values for each study.



These standard deviations generally are quite small (note the log scale)–this suggests most standard errors are precisely estimated, or at least that they do not change much as we permute treatment assignment.

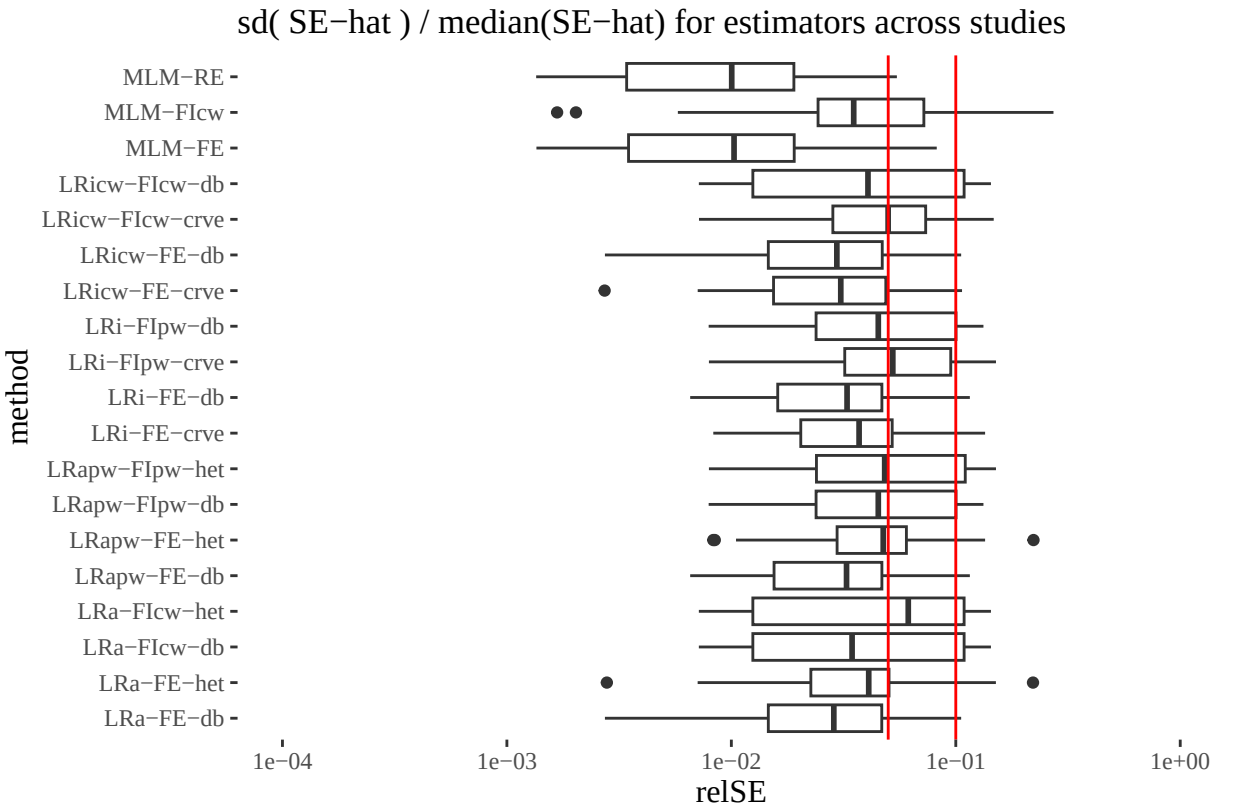
We can look at these relative to the median SE, to turn them all into ratios.



Interestingly, many of the $sd(\widehat{SE})$ values are extremely stable; the standard deviations are virtually zero. The permutations do not change much of anything. The most stable we see are the following:

UniqueID	method	SE	ESEhat	sdSEhat	relSE
TRIADScale-up_rema2	LRicw-FIcw-crve	0.265	0.006	0	0.000
TRIADScale-up_rema2	LRi-FIpw-crve	0.270	0.005	0	0.000
jg_crt_19	MLM-RE	0.042	0.051	0	0.001
jg_crt_19	MLM-FE	0.042	0.051	0	0.001
jg_crt_29	MLM-RE	0.022	0.021	0	0.002
jg_crt_29	MLM-FE	0.022	0.021	0	0.002
jg_crt_29	MLM-FIcw	0.022	0.021	0	0.002
jg_crt_18	MLM-RE	0.033	0.039	0	0.002
jg_crt_18	MLM-FE	0.033	0.039	0	0.002
jg_crt_30	MLM-FIcw	0.018	0.019	0	0.002
jg_crt_30	MLM-RE	0.019	0.019	0	0.002
jg_crt_30	MLM-FE	0.019	0.019	0	0.002
TennStar_gktmathss	MLM-FE	0.039	0.048	0	0.002
jg_crt_17	MLM-RE	0.048	0.035	0	0.002
jg_crt_17	MLM-FE	0.049	0.035	0	0.002
TennStar_gktmathss	MLM-RE	0.039	0.047	0	0.002
jg_crt_34	MLM-RE	0.082	0.075	0	0.002
jg_crt_34	MLM-FE	0.083	0.075	0	0.003
jg_crt_16	LRicw-FE-crve	0.084	0.076	0	0.003
jg_crt_16	LRicw-FE-db	0.084	0.077	0	0.003

We can see if some estimators are more stable than others:



Indeed some are. Note that if a box edge hits the 0.10 line, then more than 25% of the studies have a $sd(\widehat{SE})$ that is more than 10% of the median true SE.

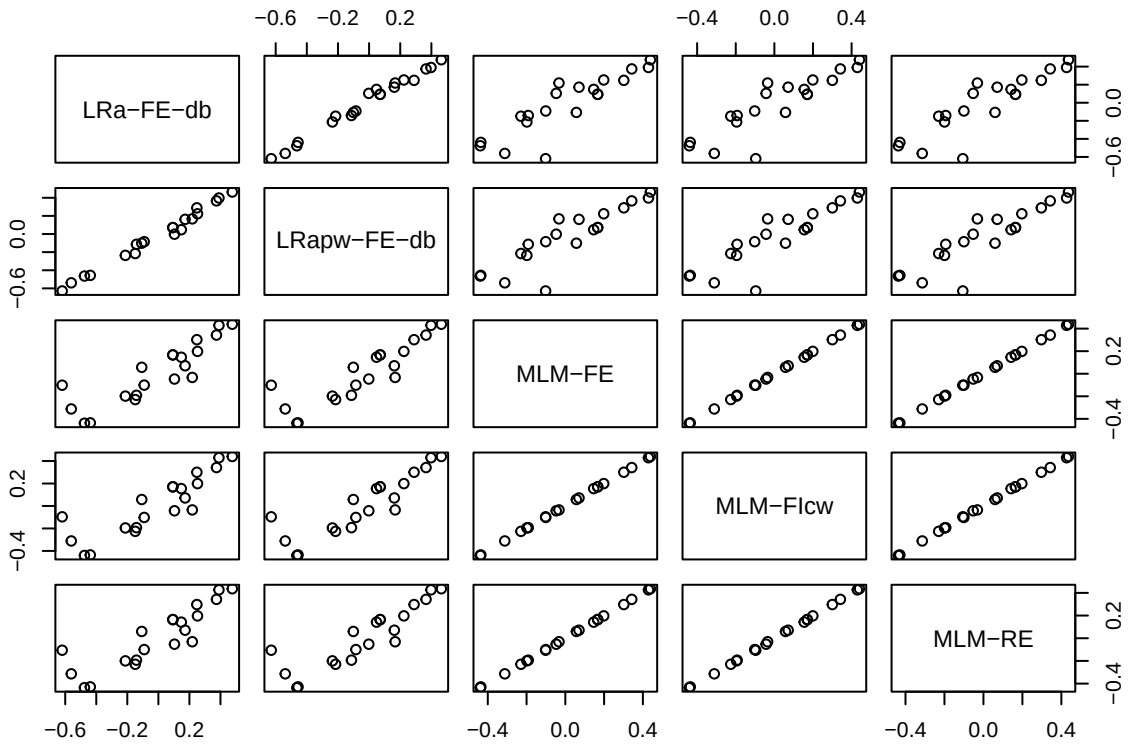
Concern: Permutation does not change the within cluster variation, or block variation. It just reassigns who is tx and who co. This means many estimated SEs are extremely stable some across permutations, and thus we are getting a false sense of SE estimatator stability. We need to use a separate simulation that generates new outcome data given the fingerprint to really understand how stable the SE-hat estimators are.

Digging into TRIADScale-up_rema2

We can look at `TRIADScale-up_rema2`.

How do the point estimates correlate?

Here we look at how point estimates of the different methods correlate with other point estimates across the permutations. Each point corresponds to two estimators applied to a single permutation. We are looking at all of the estimators that are unique, several estimators were dropped because they are not defined for this data set.

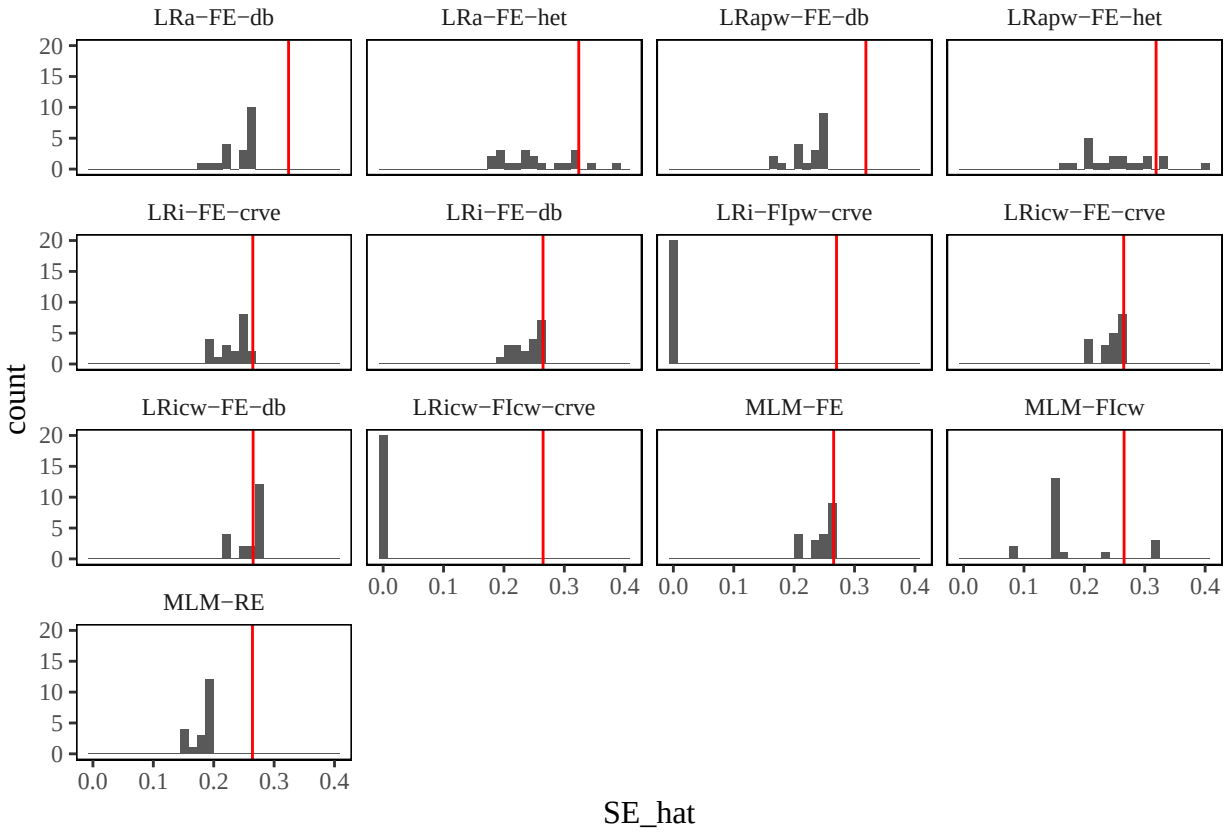


How do the estimated SEs look?

We look at the estimated standard errors across the simulations. We are seeing some estimators reliabily give very different standard error estimates.

It looks like the randomness of permutation does not translate to a wide range of standard error estimates. The red lines showed the true standard error across the permutations. Under various permutations, some of the estimators are giving systematically wrong answers. This would suggest that the permutation simulation is not capturing the correct randomness in some cases: all permutations are producing broken SE-hats.

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



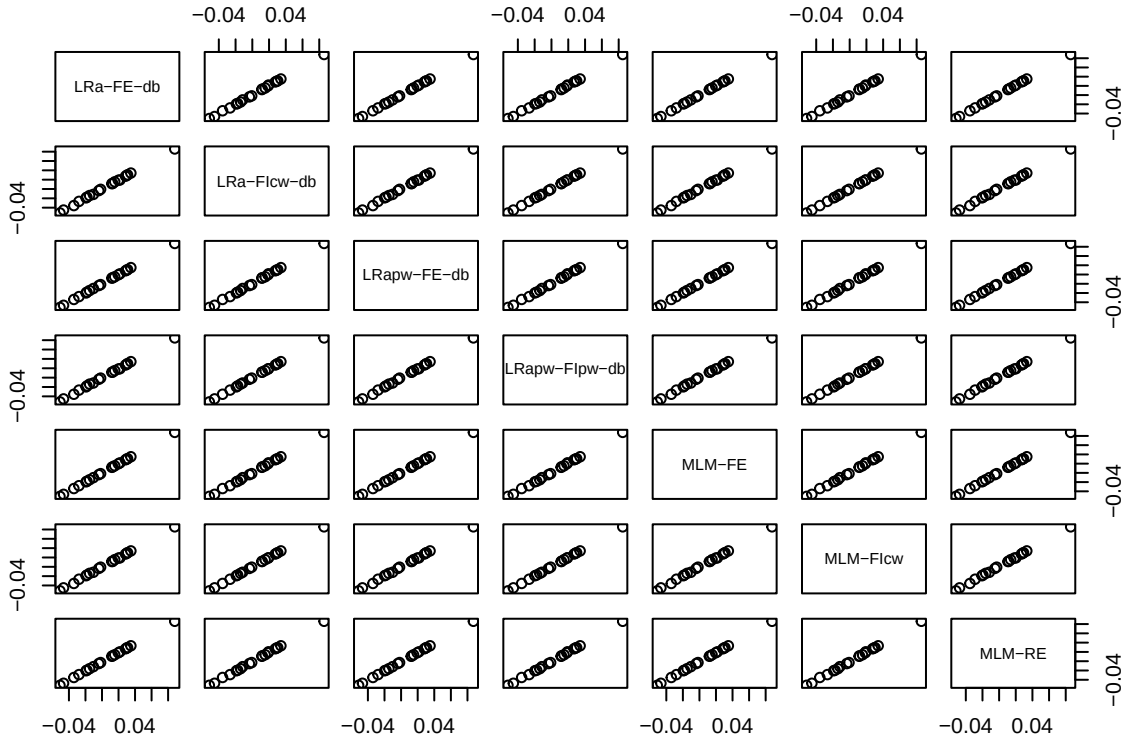
And how do the estimators perform?

method	SE	ESEhat	sdSEhat
MLM-FIcw	0.266	0.172	0.072
LRapw-FE-het	0.319	0.254	0.059
LRa-FE-het	0.324	0.257	0.059
LRapw-FE-db	0.319	0.224	0.029
LRa-FE-db	0.324	0.241	0.027
LRi-FE-crve	0.265	0.232	0.024
LRi-FE-db	0.265	0.238	0.024
LRicw-FE-db	0.265	0.262	0.023
LRicw-FE-crve	0.265	0.243	0.022
MLM-FE	0.265	0.243	0.022
MLM-RE	0.264	0.183	0.015
LRi-FIpw-crve	0.270	0.005	0.000
LRicw-FIcw-crve	0.265	0.006	0.000
LRa-FIcw-db	NA	NA	NA
LRa-FIcw-het	NA	NA	NA
LRapw-FIpw-db	NA	NA	NA
LRapw-FIpw-het	NA	NA	NA
LRi-FIpw-db	NA	NA	NA
LRicw-FIcw-db	NA	NA	NA

Digging into jg_crt_63

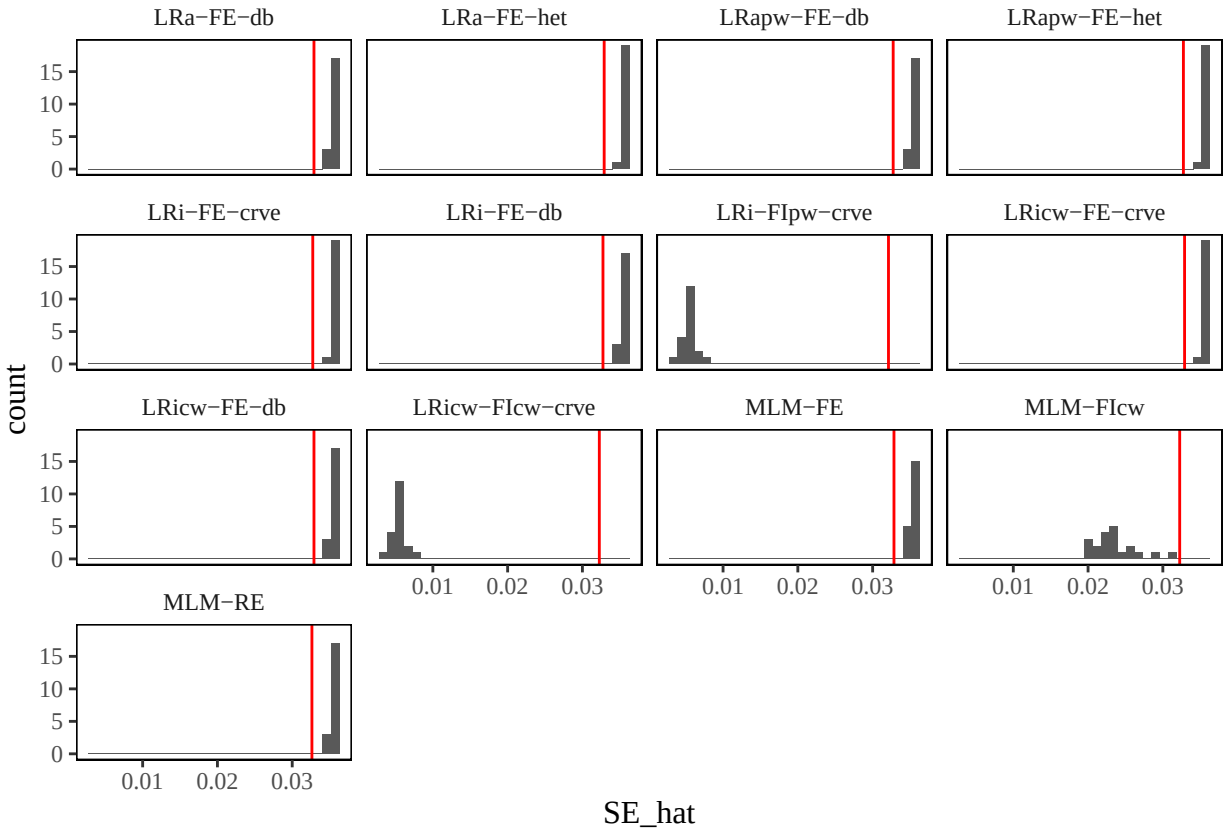
We can look at `jg_crt_63`, the other "nearly a matched pair" scenario:

How do the point estimates correlate?



How do the estimated SEs correlate?

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



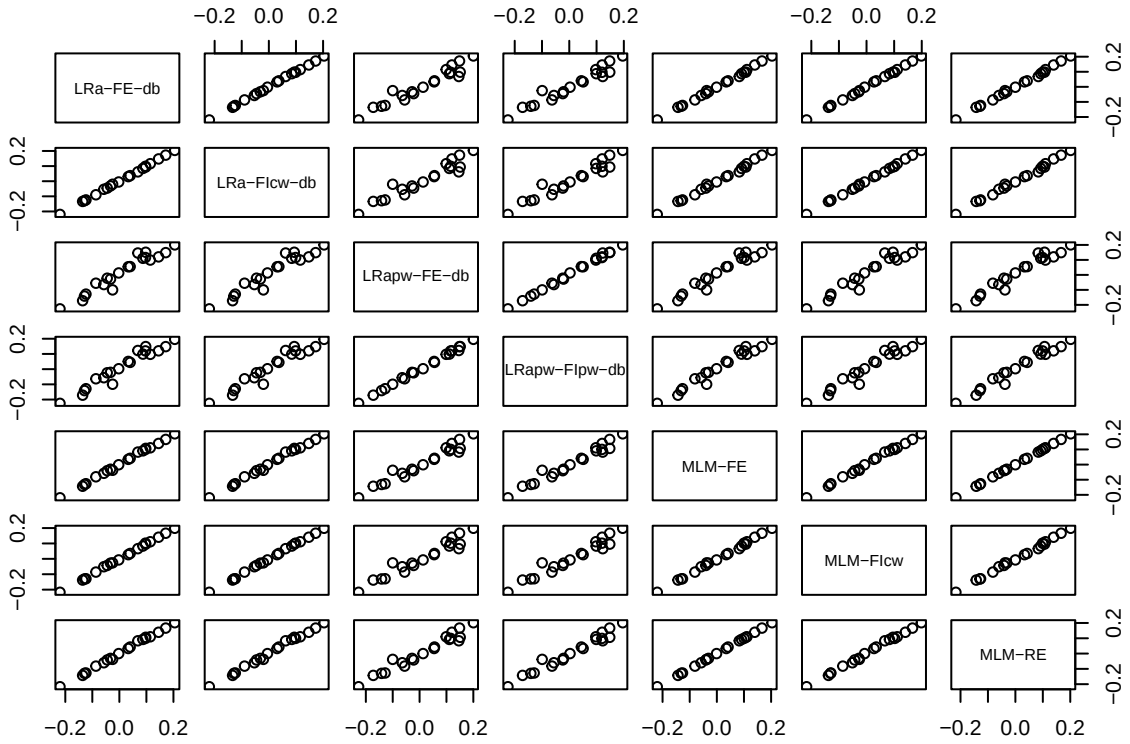
And how do the estimators perform?

method	SE	ESEhat	sdSEhat
MLM-FIcw	0.032	0.024	0.003
LRi-FIpw-crve	0.032	0.005	0.001
LRicw-FIcw-crve	0.032	0.005	0.001
LRa-FE-db	0.033	0.035	0.000
LRicw-FE-db	0.033	0.035	0.000
LRa-FE-het	0.033	0.036	0.000
LRicw-FE-crve	0.033	0.036	0.000
LRi-FE-db	0.033	0.035	0.000
LRapw-FE-db	0.033	0.035	0.000
LRi-FE-crve	0.033	0.036	0.000
LRapw-FE-het	0.033	0.036	0.000
MLM-RE	0.033	0.035	0.000
MLM-FE	0.033	0.035	0.000
LRa-FIcw-db	0.032	NA	NA
LRa-FIcw-het	0.032	NA	NA
LRapw-FIpw-db	0.032	NA	NA
LRapw-FIpw-het	0.032	NA	NA
LRi-FIpw-db	0.032	NA	NA
LRicw-FIcw-db	0.032	NA	NA

Digging into jg_crt_23

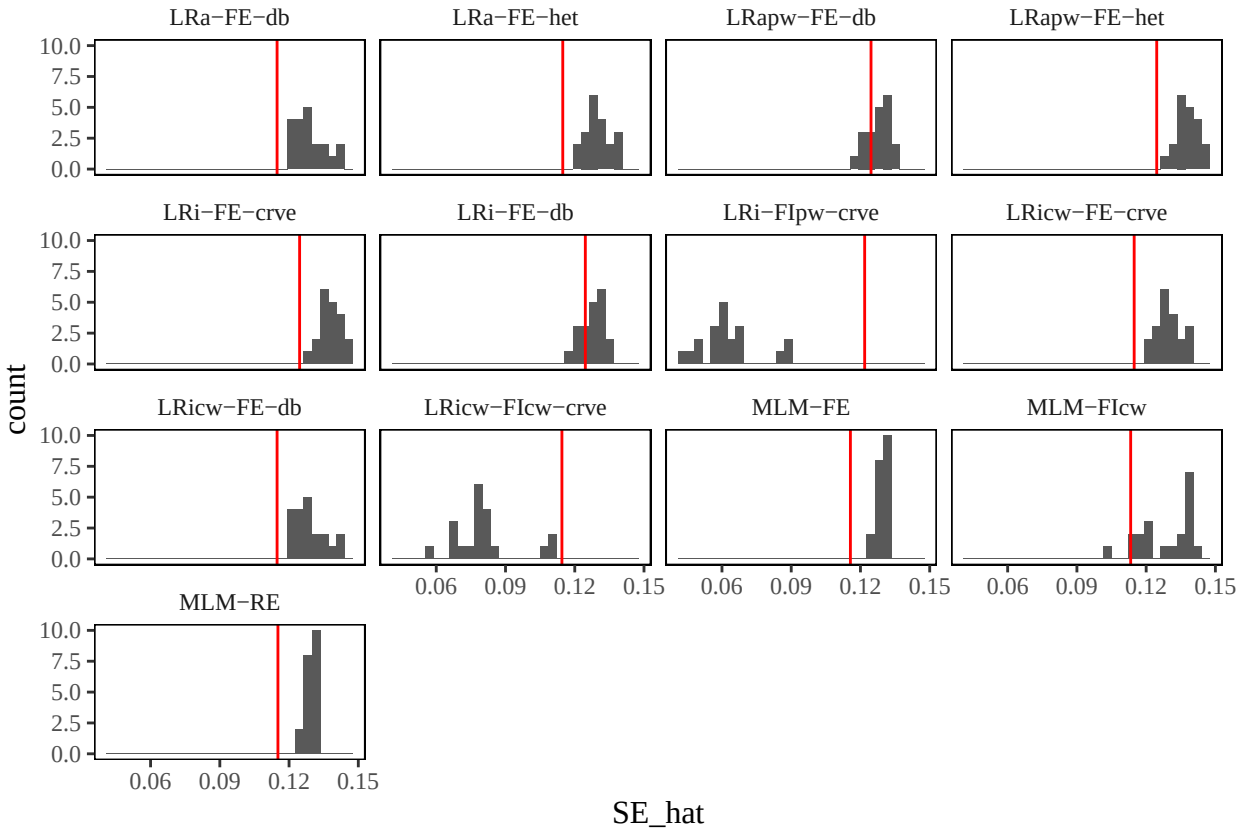
We can look at `jg_crt_23`, one of the studies that was not remarkable in terms of poor performance of anything.

How do the point estimates correlate?



How do the estimated SEs correlate?

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



And how do the estimators perform?

method	SE	ESEhat	sdSEhat
LRicw-FIcw-crve	0.114	0.080	0.014
LRi-FIpw-crve	0.122	0.063	0.012
MLM-FIcw	0.113	0.128	0.012
LRicw-FE-db	0.115	0.129	0.007
LRa-FE-db	0.115	0.129	0.007
LRicw-FE-crve	0.115	0.130	0.006
LRa-FE-het	0.115	0.130	0.005
LRapw-FE-het	0.125	0.138	0.005
LRi-FE-crve	0.125	0.138	0.005
LRi-FE-db	0.125	0.128	0.005
LRapw-FE-db	0.125	0.128	0.005
MLM-FE	0.116	0.129	0.002
MLM-RE	0.115	0.129	0.002
LRa-FIcw-db	0.114	NA	NA
LRa-FIcw-het	0.114	NA	NA
LRapw-FIpw-db	0.122	NA	NA
LRapw-FIpw-het	0.122	NA	NA
LRi-FIpw-db	0.122	NA	NA
LRicw-FIcw-db	0.114	NA	NA

We see the general trneds of systematically high (or low) estimated SEs with little variation. Permutation is a poor way of assessing instability of the SE-hats.